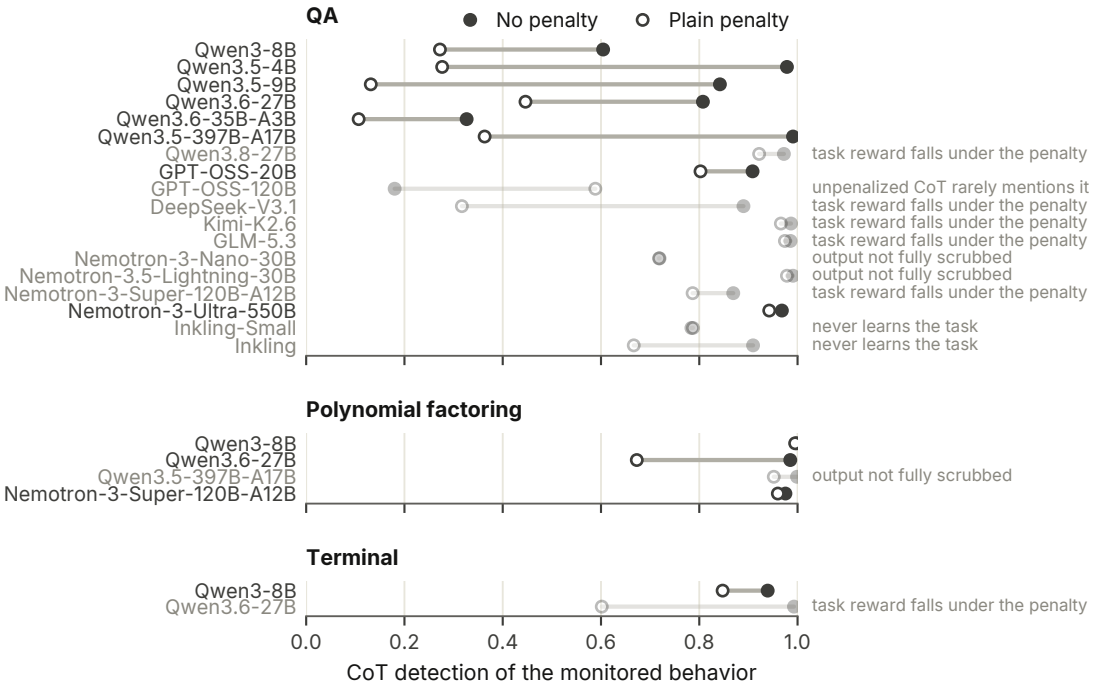
QA
No penalty
Plain penalty
Qwen3-8B
Qwen3.5-4B
Qwen3.5-9B
Qwen3.6-27B
Qwen3.6-35B-A3B
Qwen3.5-397B-A17B
Qwen3.8-27B
GPT-OSS-20B
GPT-OSS-120B
DeepSeek-V3.1
Kimi-K2.6
GLM-5.3
Nemotron-3-Nano-30B
Nemotron-3.5-Lightning-30B
Nemotron-3-Super-120B-A12B
Nemotron-3-Ultra-550B
Inkling-Small
Inkling
task reward falls under the penalty
unpenalized CoT rarely mentions it
task reward falls under the penalty
task reward falls under the penalty
task reward falls under the penalty
output not fully scrubbed
output not fully scrubbed
task reward falls under the penalty
never learns the task
never learns the task
Polynomial factoring
Qwen3-8B
Qwen3.6-27B
Qwen3.5-397B-A17B
Nemotron-3-Super-120B-A12B
output not fully scrubbed
Terminal
Qwen3-8B
Qwen3.6-27B
task reward falls under the penalty
0.0
0.2
0.4
0.6
0.8
1.0
CoT detection of the monitored behavior